Marketing Funnel & Conversion Performance Analysis

1. Project Overview

- This project analyzes marketing campaign data to understand customer behaviour across the conversion funnel from initial website visits to final campaign response.
- The goal is to identify where users drop off, evaluate the effectiveness of different marketing channels, and uncover factors influencing customer conversion.
- Using real-world data from Maven Analytics, the analysis focuses on key stages of the funnel: visitors, leads, and customers. By examining customer demographics, purchasing patterns, and campaign interactions, the project provides insights into improving marketing strategies and increasing conversion rates.
- The findings from this analysis help businesses optimize their marketing efforts, target the right audience segments, and enhance overall customer engagement and revenue generation.

1.1 Project Objective

- To analyze customer behaviour across the marketing funnel (Visitors → Leads → Customers).
- To identify stages where users drop off in the conversion process.
- To evaluate the performance of different marketing channels.
- To understand the impact of customer demographics on conversion.
- To measure overall conversion rates and campaign effectiveness.
- To provide actionable insights to improve customer conversion.
- To suggest strategies for optimizing marketing efforts and increasing revenue.

2. Data Sources

- The dataset used in this project is sourced from https://mavenanalytics.io/data-playground/marketing-campaign-results.
- It is a publicly available dataset provided for practicing data analysis and visualization.
- **Domain:** This project falls under **Marketing Analytics and Business Intelligence**, focusing on customer behavior, campaign performance, and conversion optimization.

3. Problem Statement

- Businesses lack clear visibility into the customer conversion funnel
- Difficulty in identifying where users drop off (visits → purchases → conversions)
- Uncertainty about which marketing channels perform best
- Limited understanding of factors influencing customer conversion
- Inefficient targeting and marketing strategies due to lack of insights
- Need for data-driven analysis to improve conversion rates and campaign effectiveness

4. Original Dataset Attributes (Column /Features) Details

Column Name	Description	Data Type
ID	Unique customer identifier	Integer
Year_Birth	Year of birth of the customer	Integer
Education	Education level	Categorical (Text)
Marital_Status	Marital status	Categorical (Text)
Income	Annual income of the customer	Numeric (Decimal)
Kidhome	Number of children in household	Integer
Teenhome	Number of teenagers in household	Integer
Dt_Customer	Date of customer enrollment	Date
Recency	Days since last purchase	Integer

Product Spending

Column Name	Description	Data Type
MntWines	Amount spent on wine	Numeric
MntFruits	Amount spent on fruits	Numeric
MntMeatProducts	Amount spent on meat products	Numeric
MntFishProducts	Amount spent on fish products	Numeric
MntSweetProducts	Amount spent on sweets	Numeric
MntGoldProds	Amount spent on gold products	Numeric

Purchase Behavior

Column Name	Description	Data Type
NumDealsPurchases	Number of purchases made with discount	Integer
NumWebPurchases	Number of web purchases	Integer
NumCatalogPurchases	Number of catalog purchases	Integer
NumStorePurchases	Number of store purchases	Integer
NumWebVisitsMonth	Number of website visits per month	Integer

Campaign Interaction

Column Name	Description	Data Type
AcceptedCmp1	Accepted campaign 1 (1/0)	Integer
AcceptedCmp2	Accepted campaign 2 (1/0)	Integer
AcceptedCmp3	Accepted campaign 3 (1/0)	Integer
AcceptedCmp4	Accepted campaign 4 (1/0)	Integer
AcceptedCmp5	Accepted campaign 5 (1/0)	Integer
Response	Accepted last campaign (1 = Yes, 0 = No)	Integer

Other

Column Name	Description	Data Type
Complain	Complaint status (1 = Yes, 0 = No)	Integer
Country	Customer location	Categorical (Text)

6. Tools & Technologies

- **Microsoft Office Professional Plus 2019(64-Bit):** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

7. Data Pre-Processing (Excel / Power Query)

- **Initial Data Understanding & Data Cleaning:** The dataset was first explored to understand its structure and quality before performing analysis.
  - Identified total number of rows (2241) and columns (25) in the dataset.
  - Reviewed each column to understand its meaning and relevance.
  - Checked data types of all columns (numeric, categorical, date)
  - Examined each column individually for missing or null values (e.g., Income column has blank values).
  - Handled missing values appropriately (e.g., Income column)
  - Renamed inconsistent or unclear values (e.g., Renamed “2n Cycle” from education column to “Diploma” using find and replace)
  - Ensured column names and values are clear and consistent

- No completely irrelevant columns were found; however, some columns were treated as optional based on analysis requirements
- Verified overall data quality before proceeding to cleaning and transformation

- This initial step helped in understanding the dataset and preparing it for further data cleaning and analysis.

Descriptive statistics of Income	
Mean	52247.25
Median	51381.5
Mode	7500
Standard Deviation	25173.07
Kurtosis	159.64
Skewness	6.76
Minimum	1730
Maximum	666666
Count	2216

- Performed descriptive statistical analysis using Excel Analysis ToolPak on numerical column Income.
- Calculated key metrics including mean, median, standard deviation, minimum, maximum, and skewness
- Observed that the income data has a wide range with significant variation between minimum and maximum values
- Identified high standard deviation, indicating large variability in customer income levels
- Detected high positive skewness, suggesting the presence of extreme values (outliers)
- Noted that a small number of customers have very high income compared to the majority
- Confirmed that the data distribution is not perfectly normal and is slightly right-skewed
- Although mean and median values are relatively close, further analysis showed high skewness in the data
- Presence of extreme maximum values indicates outliers in the income distribution
- Mean is sensitive to outliers and can give a distorted representation of central tendency
- Median is more robust and not affected by extreme values
- Therefore, median was chosen to handle missing values as it provides a more accurate and reliable representation of typical customer income.

- Replaced blank values of the income column with the median value using the below EXCEL formula.

```
=IF([@Income]="",Analysis_Tool_Pak!$B$5,[@Income])
```

- Cleaned the Marital_Status column by handling invalid and inconsistent values
- Replaced non-meaningful categories such as “YOLO” and “Absurd” with “Single”
- Ensured consistent and meaningful categories for better analysis and segmentation

```
=IFS([@[Marital_Status]]="Alone","Single",
[@[Marital_Status]]="Absurd","Single",
[@[Marital_Status]]="YOLO","Single",
[@[Marital_Status]]="Together","Married",TRUE,[@[Marital_Status]])
```

- Retained Kidhome and Teenhome as separate columns to preserve detailed household information
- Created a new column “TotalChildren” by combining both features using below formula to represent total dependents.

```
=[@Kidhome]+[@Teenhome]
```

- This enabled both detailed and aggregated analysis of customer behavior based on family size.
- Dataset also includes product-wise spending columns representing the amount spent by customers across different product categories over the last two years
- These features were used to analyze customer purchasing behavior and identify high-value customers.
- Created a new column **“Total Engagement”** using the below excel formula by combining all purchase interactions including web, store, catalog, and discount-based purchases.

```
=[@NumDealsPurchases]+[@NumWebPurchases]+[@NumCatalogPurchases]+[@NumStorePurchases]
```

- This feature was used to measure overall customer engagement and support funnel analysis.

- In the original dataset, product-related columns were stored separately:Wines, Fish, Meat, Gold, Sweet and Fruits
- This structure is called **wide format**, which is not suitable for analysis.
- Using **Power Query (Unpivot Columns)**-Selected all product columns and applied **Unpivot Columns.**
- Unpivoting columns created two columns: Product and Amount_Spent_on_Product
- To follow data modeling best practices, created a separate product dimension table called **Dim_Product**.
- Dim_Product has 2 columns named Product and Product_ID.
- Product_ID column filled using the below excel formula.

```
="P" & TEXT(ROW([@Product])-1,"00")
```

- To connect the main dataset with dim_product, added a new column named product-ID in the main data set and filled the values using the below vlookup formula.

```
=VLOOKUP([@Product],Table6,2,FALSE)
```

- The dataset contained multiple campaign-related columns: AcceptedCmp1, AcceptedCmp2, AcceptedCmp3, AcceptedCmp4, AcceptedCmp5 and Response. These columns represent **customer responses to different marketing campaigns**, but were stored in a **wide format**, making analysis difficult.
- To normalize the dataset, used **Unpivot Columns** in Power Query on all campaign-related fields.
- A separate dimension table "**Dim_Campaign"** was created to store campaign details.
- A campaign_id column was created in the marketing dataset using below formula by mapping campaign names with the dimension table.

```
=VLOOKUP([@Attribute],Table8,2,FALSE)
```

- Unpivoting was performed to convert wide-format data into long-format, enabling better aggregation, filtering, and compatibility with dimensional modeling.

Fact Table: fact_marketing_data

The transformed dataset was structured as a fact table containing transactional and campaign-level information.

Columns include:

- customer_id (Foreign Key)
- product_id (Foreign Key)
- campaign_id (Foreign Key)
- accepted (Campaign response: 1/0)
- Purchase-related metrics (web, store, catalog purchases)
- Engagement metrics (web visits, deals purchases)

- The "Response" column was treated as the final stage of the funnel and renamed to "Final Campaign" to maintain consistency in campaign stages.

8. Data Modeling:

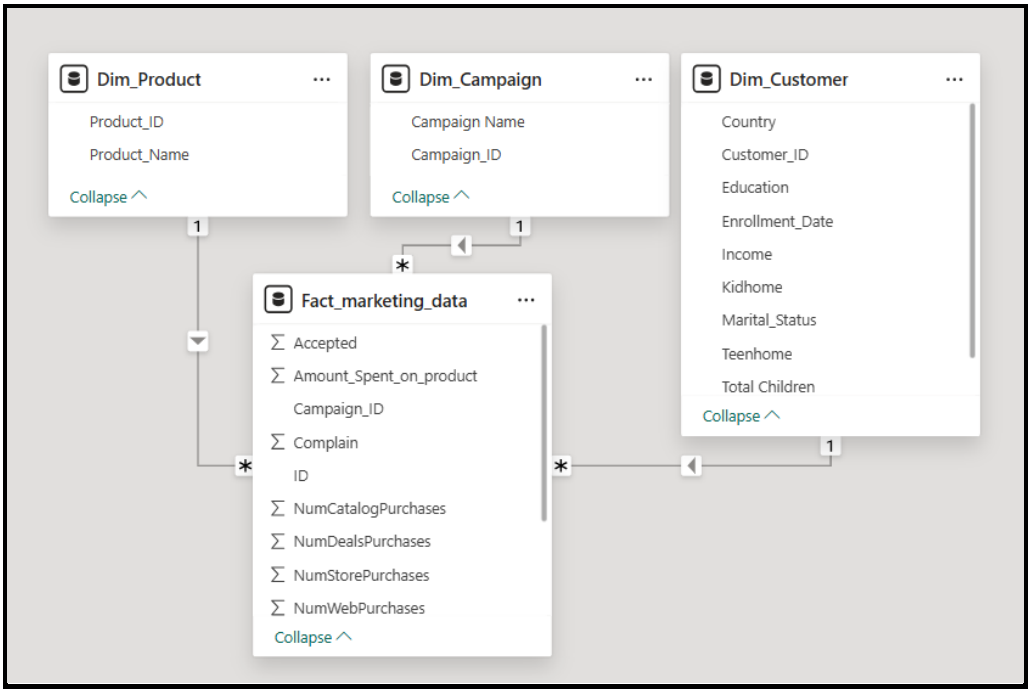
- A star schema was implemented for the marketing funnel dataset. Separate dimension tables were created for customers, products, and campaigns. The main marketing table was transformed into a fact table by removing descriptive columns and retaining only foreign keys (customer_id, product_id, campaign_id) along with relevant numerical and funnel-related measures.
- This modeling approach reduced redundancy, improved data consistency, and enabled efficient funnel analysis in Power BI.

Relationships:

- The following relationships were established:
  - Dim_customer (1) → (Many) fact_marketing_data
  - Dim_product (1) → (Many) fact_marketing_data
  - Dim_campaign (1) → (Many) fact_marketing_data

Relationship Type:

- One-to-Many
- Single direction (Dimension → Fact)



- The dataset was successfully transformed and modeled into a star schema, enabling efficient analysis of the marketing funnel.
- By structuring campaign interactions as funnel stages and integrating customer and product dimensions, the model supports in-depth analysis of conversion behavior.
- This approach allows businesses to identify drop-off points, optimize campaigns, and improve overall marketing effectiveness.

9. DAX Measures:

- **Total Customers:** Calculates the total number of unique customers in the dataset.

```
No.of Customers = DISTINCTCOUNT(Fact_marketing_data[ID])
```

- **Conversions:** Calculates the number of customers who accepted the campaign.

```
Conversions = CALCULATE(
    DISTINCTCOUNT(fact_marketing_data[ID]),
    fact_marketing_data[Accepted] = 1 )
```

- **Conversion Rate:** Represents the percentage of customers who converted.

```
Conversion Rate = Divide([Conversions],[No.of Customers])
```

- **Drop-offs:** Calculates the number of customers who did not convert.

```
Drop off Count = [No.of Customers]-[Conversions]
```

- **Drop-off Rate:** Represents the percentage of customers who dropped off in the funnel.

```
Drop off Rate = 1-[Conversion Rate]
```

- **Total Spend:** Calculates the total amount spent by customers across all products.

```
Total Spend per Customer =
CALCULATE(
    SUM(fact_marketing_data[Amount_Spent_on_product]),
    ALLEXCEPT(fact_marketing_data, fact_marketing_data[ID])
)
```

- **Average Spend:** Calculates the average amount spent per customer

```
Average Spend =
DIVIDE(
    [Total Spend per Customer],
    DISTINCTCOUNT(fact_marketing_data[ID])
)
```

- **Channel Conversions:** Calculates the number of converted customers based on the selected purchase channel (Web, Store, or Catalog).

```
Channel Conversions =
SWITCH(
    SELECTEDVALUE('Dim-Channel'[Channel]),
    "Web",
        CALCULATE(
            DISTINCTCOUNT(fact_marketing_data[ID]),
            fact_marketing_data[Accepted] = 1,
            fact_marketing_data[NumWebPurchases] > 0
        ),
    "Store",
        CALCULATE(
            DISTINCTCOUNT(fact_marketing_data[ID]),
            fact_marketing_data[Accepted] = 1,
            fact_marketing_data[NumStorePurchases] > 0
        ),
    "Catalog",
        CALCULATE(
            DISTINCTCOUNT(fact_marketing_data[ID]),
            fact_marketing_data[Accepted] = 1,
            fact_marketing_data[NumCatalogPurchases] > 0
        )
)
```

- **Channel Purchases:** Calculates total number of purchases based on the selected channel (Web, Store, or Catalog).

```
Channel Purchases =
SWITCH(
    SELECTEDVALUE('Dim-Channel'[Channel]),
    "Web", SUM(fact_marketing_data[NumWebPurchases]),
    "Store", SUM(fact_marketing_data[NumStorePurchases]),
    "Catalog", SUM(fact_marketing_data[NumCatalogPurchases])
)
```

- **Product Conversion Rate:** Calculates the conversion rate for customers who purchased a specific product.

```
Product Conversion Rate =
DIVIDE(
    CALCULATE(
        DISTINCTCOUNT(fact_marketing_data[ID]),
        fact_marketing_data[Accepted] = 1,
        fact_marketing_data[Amount_Spent_on_product] > 0
    ),
    CALCULATE(
        DISTINCTCOUNT(fact_marketing_data[ID]),
        fact_marketing_data[Amount_Spent_on_product] > 0
    )
)
```

- **Complaint Rate:** Calculates the percentage of customers who have raised complaints.

```
Complaint Rate =
DIVIDE(
    COUNTROWS(FILTER(fact_marketing_data, fact_marketing_data[Complain] = 1)),
    COUNTROWS(fact_marketing_data)
)
```

- **Average Web Purchases:** Calculates the average number of web purchases made by customers.

```
Avg Web Purchases =
AVERAGE(fact_marketing_data[NumWebPurchases])
```

- These measures use dynamic filtering and conditional logic to enable flexible analysis across channels and products. By leveraging functions such as SWITCH, SELECTEDVALUE, and CALCULATE, the measures adapt based on user interaction, supporting interactive and insightful dashboard exploration.

10. Calculated Columns:

- **Age:** Calculates the current age of customers based on their year of birth.

```
Age = YEAR(TODAY()) - Dim_Customer[Year_Birth]
```

- **Age Group:** Classifies customers into different age segments such as Young Adults, Adults, Middle Age, and Senior.

```
Age Group =
SWITCH(
    TRUE(),
    [Age] < 25, "Young Adults",
    [Age] < 40, "Adults",
    [Age] < 55, "Middle Age",
    "Senior"
)
```

- **Income Category:** Segments customers into income groups based on their annual income.

```
Income Category =
SWITCH(
    TRUE(),
    Dim_Customer[Income] < 25000, "Very Low",
    Dim_Customer[Income] < 50000, "Low",
    Dim_Customer[Income] < 80000, "Medium",
    Dim_Customer[Income] < 120000, "High",
    "Very High"
)
```

- **Total Spend:** Calculates the total amount spent by each customer across all products.

```
Total Spend =
CALCULATE(
    SUM(fact_marketing_data[Amount_Spent_on_product])
)
```

- **Spend Group:** Categorizes customers based on their total spending into groups such as Low, Medium, High and Very High.

```
Spend Group =
SWITCH(
    TRUE(),
    Dim_Customer[Total Spend] <= 500, "Low",
    Dim_Customer[Total Spend] < 3000, "Medium",
    Dim_Customer[Total Spend] < 7000, "High",
    "Very High"
)
```

- **Recency Group:** Groups customers based on how recently they made a purchase.

```
Recency Group =
SWITCH(
    TRUE(),
    fact_marketing_data[Recency] <= 30, "Recent",
    fact_marketing_data[Recency] <= 90, "Moderate",
    fact_marketing_data[Recency] <= 180, "Old",
    "Very Old"
)
```

- **Web Visit Group:** Segments customers based on the number of website visits per month.

```
Web Visit Group =
SWITCH(
    TRUE(),
    fact_marketing_data[NumWebVisitsMonth] <= 2, "Low",
    fact_marketing_data[NumWebVisitsMonth] <= 5, "Medium",
    fact_marketing_data[NumWebVisitsMonth] <= 10, "High",
    "Very High"
)
```

- **Deal Purchase Group:** Segments customers based on number of purchases made using discounts.

```
Deal Purchase Group =
SWITCH(
    TRUE(),
    fact_marketing_data[NumDealsPurchases] = 0, "No Deals",
    fact_marketing_data[NumDealsPurchases] <= 3, "Low",
    fact_marketing_data[NumDealsPurchases] <= 7, "Medium",
    fact_marketing_data[NumDealsPurchases] <= 12, "High",
    "Very High"
)
```

- These calculated columns were created to enhance data segmentation and support detailed analysis of customer demographics, spending behaviour, and conversion patterns. They enable more meaningful grouping and improve the interpretability of dashboard insights.

11. Analysis and Visualizations with Insights (Power BI)

- The dashboard includes multiple visual elements designed to analyse marketing funnel performance, customer segmentation, and behavioural patterns influencing conversion:
    - **KPI Cards**: Display key metrics such as Total Customers, Total Conversions, Conversion Rate, Drop-offs, Drop-off Rate and Complaint Rate, providing a quick overview of overall funnel performance.
    - **Bar Charts:** Represent conversions and drop-off rates across different campaign stages, helping identify critical points where customer attrition occurs.
    - **Combo Chart**: Shows the number of customers and conversion rate by enrollment year, enabling comparison between customer acquisition trends and performance over time.
    - **Bar Chart (Channel Analysis)**: Displays total purchases by channel (Web, Store, Catalog), helping evaluate channel-wise contribution to overall sales.
    - **Donut Chart**: Represents conversion share by channel, highlighting the distribution of customer engagement across different purchase channels.
    - **Bar Charts (Segmentation Analysis)**: Used to analyze conversion rates across income categories and education levels, helping identify high-performing customer segments.
    - **Pie and Donut Charts**: Show conversion patterns across age groups and marital status, enabling demographic-based analysis.

    - **Map Visual**: Displays country-wise conversion rates, providing geographic insights into customer behavior.
    - **Treemap**: Represents drop-off rate by country, helping identify regions with higher customer attrition.
    - **Matrix Visual**: Displays customer distribution across total children and spend groups, supporting multi-dimensional analysis of family size and spending behavior.
        - The matrix helps validate insights by combining customer count with spend group distribution, ensuring that conclusions are supported by sufficient data volume.
  - **Stacked Bar Chart (Product Analysis)**: Shows product-wise conversion rate across spend groups, highlighting product performance across different customer segments.
  - **Column Chart (Spending Analysis)**: Displays total amount spent by product category, identifying key revenue-driving products.
  - **Behavioral Analysis Charts** include:

    - Conversion Rate by Recency Group

    - Conversion Rate by Web Visit Group (with customer count)

    - Average Web Purchases by Web Visit Group

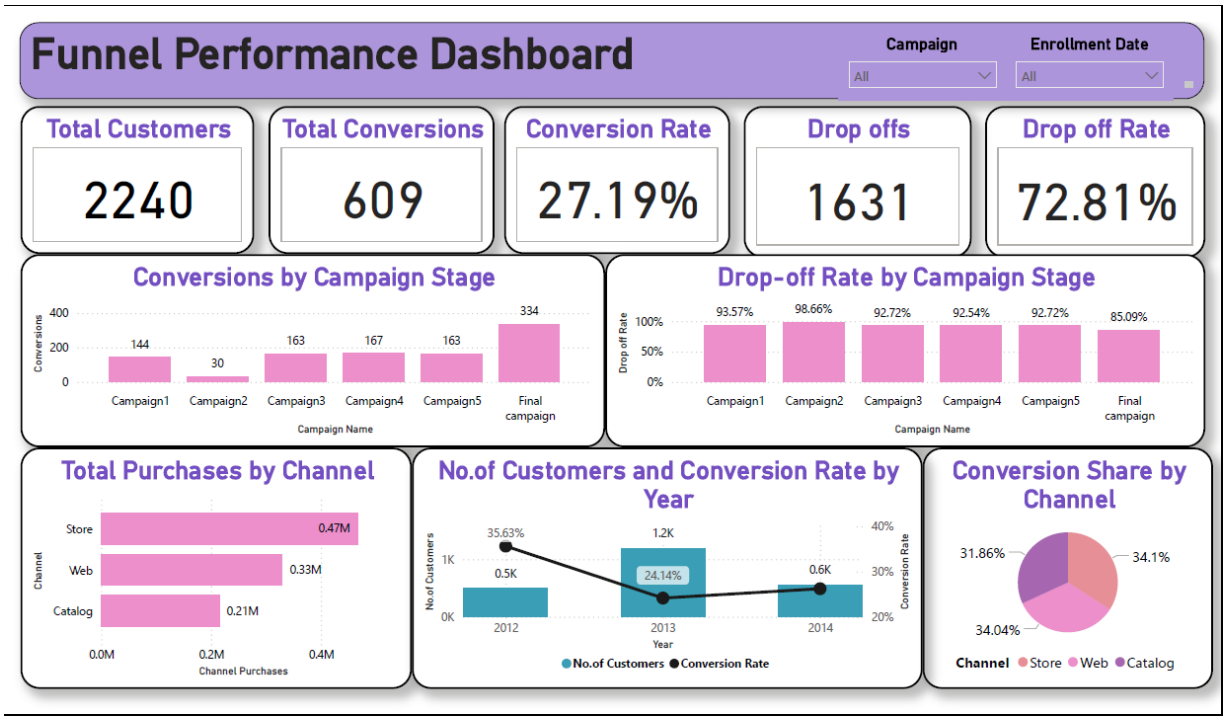
    - Conversion Rate by Complaint Status

    - Conversion Rate by Deal Purchase Group

- These visuals provide insights into how customer engagement, satisfaction, and price sensitivity influence conversion behavior.
- These visuals were selected to effectively communicate trends, comparisons, distributions, and behavioral patterns across the marketing funnel.

Consolidated Report/Dashboard:

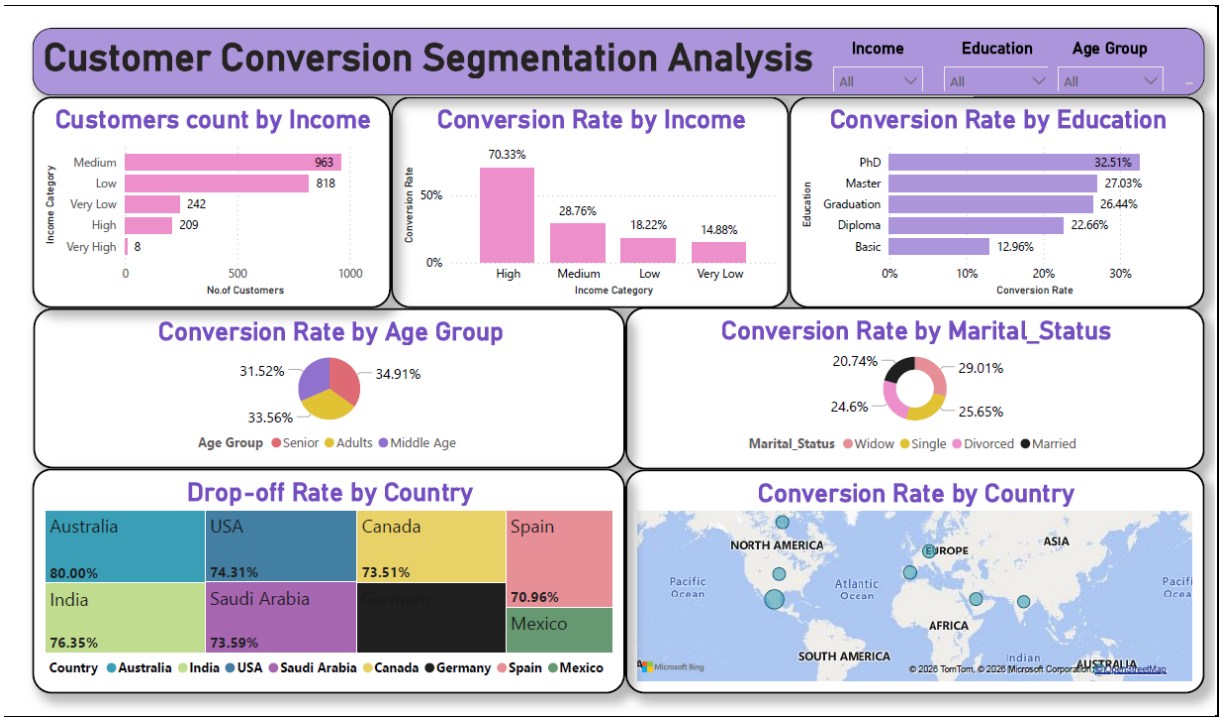
Funnel Performance Analysis



Insights:

- The overall conversion rate is 27.19%, indicating that a significant portion of customers drop off during the marketing funnel. The drop-off rate of 72.81% highlights inefficiencies in converting engaged customers into buyers.
- Campaign-level analysis shows that early campaigns generate relatively lower conversions, while the final campaign achieves the highest conversions. This suggests that customers require multiple touchpoints before making a purchase decision.
- The drop-off rate remains consistently high across initial campaign stages, indicating potential gaps in engagement, targeting, or messaging effectiveness in the early funnel.
- Customer acquisition peaked in 2013; however, conversion rate declined during this period, indicating that increased customer volume does not necessarily lead to better performance.
- Channel analysis reveals that Store and Web channels contribute more significantly to total purchases compared to Catalog, indicating stronger customer preference for direct and digital purchasing channels.

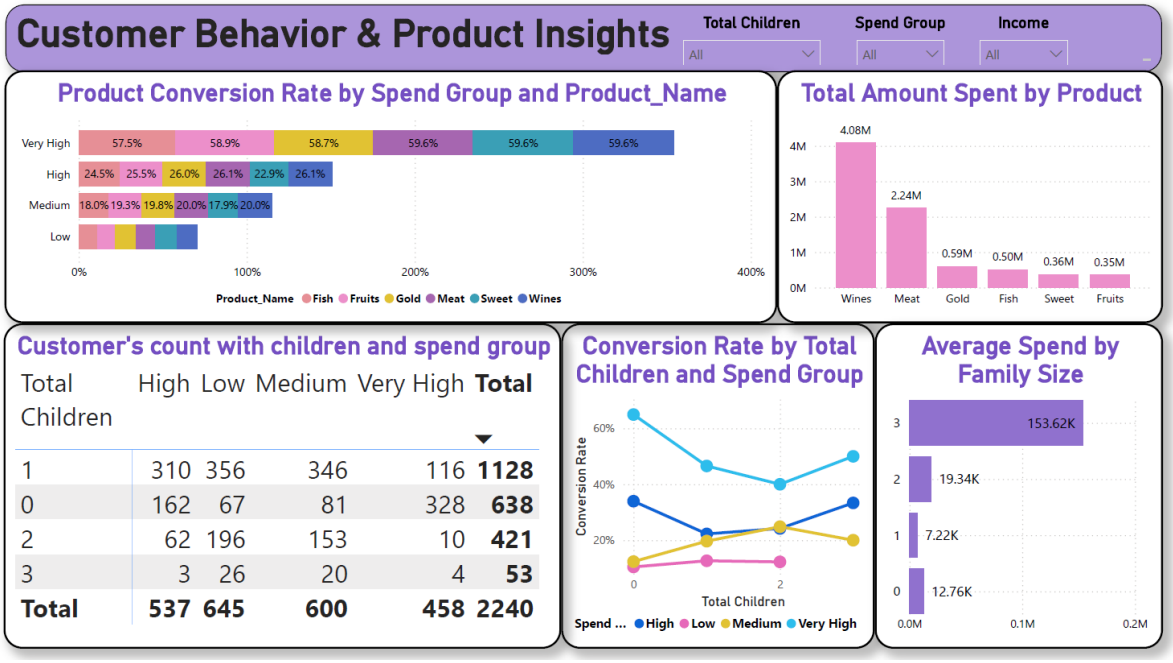
Customer Conversion Segmentation Analysis



Insights:

- Income-based analysis shows that Medium and High-income customers have higher conversion rates, making them the most valuable customer segments. Low and Very Low-income groups show significantly lower conversion rates despite having a larger customer base.
- Education analysis indicates that customers with higher education levels (PhD and Master) exhibit better conversion rates, suggesting that awareness and decision-making ability influence purchasing behavior.
- Age group analysis shows relatively balanced conversion rates across segments, indicating that age alone is not a strong differentiating factor for conversion.
- Marital status analysis reveals minor variations, with no significant impact on conversion, suggesting that this variable has limited influence on customer behavior.
- Countries like Australia, India, and the USA show higher drop-off rates, indicating lower conversion efficiency in these regions. In contrast, countries such as Mexico show comparatively lower drop-off rates, though with smaller customer volume. This suggests the need for region-specific campaign optimization to improve performance.

Customer Behavior and Product Analysis

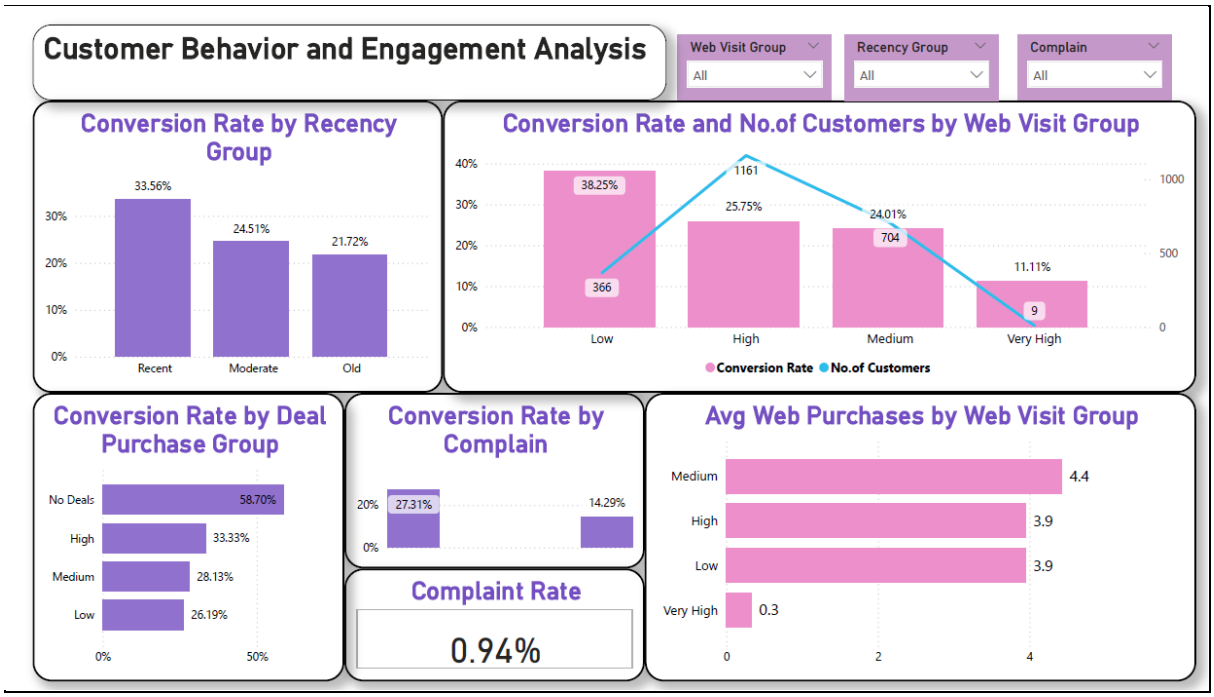


Total Children	High	Low	Medium	Very High	Total
1	310	356	346	116	**1128**
0	162	67	81	328	**638**
2	62	196	153	10	**421**
3	3	26	20	4	**53**
Total	**537**	**645**	**600**	**458**	**2240**

Insights:

- Customers with fewer children show higher conversion rates compared to those with larger family sizes, indicating that household responsibilities may impact purchasing decisions.
- Spending behavior plays a critical role in conversion. High and Very High spend groups exhibit significantly higher conversion rates, indicating that high-value customers are more responsive to campaigns.
- Product analysis shows that categories such as Wines and Meat contribute the highest total revenue, making them key drivers of business performance.
- Product conversion rates are consistently higher among high-spending customer segments, suggesting that premium customers are more likely to convert across product categories.
- The distribution of customers across segments highlights that some high-performing groups have smaller customer counts, emphasizing the need to balance both volume and performance in business decisions.

Customer Behavior and Engagement Analysis



Insights:

- Recency analysis shows that customers who made recent purchases have the highest conversion rates, while conversion declines as recency increases. This indicates that customer engagement decreases over time and highlights the importance of timely re-engagement strategies.
- Web activity analysis reveals that customers with low to moderate website visits have higher conversion rates compared to those with very high visits. This suggests that excessive browsing may indicate indecision rather than purchase intent.
- Further analysis shows that average web purchases peak at moderate engagement levels, while customers with very high web visits have significantly lower purchase activity, reinforcing the non-linear relationship between engagement and conversion.
- Complaint analysis indicates that customers who have raised complaints exhibit significantly lower conversion rates compared to those without complaints, highlighting the strong negative impact of customer dissatisfaction.
- Deal purchase behavior shows that customers who frequently rely on discounts tend to have lower conversion rates, indicating price sensitivity and reduced likelihood of conversion without incentives.

12. Final Conclusion:

- The dataset was cleaned and prepared using Excel by handling missing values, standardizing categories, and creating new derived features.
- The transformed data was then analyzed using Power BI to build interactive dashboards for funnel, segmentation, and behavioral analysis.
- Campaign analysis shows that initial campaigns experience higher drop-offs, while repeated campaign exposure improves conversion rates, highlighting the importance of continuous customer engagement.
- Customer segmentation reveals that medium and high-income groups, along with higher education levels, contribute more significantly to conversions, indicating the importance of targeted marketing.
- Behavioral analysis indicates that customers with recent activity and moderate web engagement are more likely to convert, emphasizing the role of timing and balanced interaction.
- Spending patterns show that high-value customers are more responsive to campaigns, and key product categories such as Wines and Meat drive major revenue contributions.
- Customer experience plays a critical role, as customers with complaints exhibit significantly lower conversion rates, highlighting the impact of satisfaction on business performance.
- Deal-based purchasing behavior indicates that highly price-sensitive customers are less likely to convert without incentives, suggesting the need for strategic discounting.
- Geographical analysis highlights that countries such as Australia and India show higher drop-off rates, indicating opportunities for region-specific campaign optimization.
- Overall, the analysis demonstrates that conversion is influenced by a combination of campaign strategy, customer demographics, engagement behavior, and customer experience, enabling data-driven decision-making for improved marketing effectiveness.